La Scuola d'Italia / Dr. Huson / IB Math Name:
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5.7 Classwork: Analytic Geometry (no calculator)

1. A straight line L passes through the points $A(6,0)$ and $B(0,-4)$.
   (a) Find the gradient of L. [2]

   (b) Write down the y-intercept of L. [1]

   (c) Find the equation of L in the form $ax+by+d=0$, where a, b, $d \in \mathbb{Z}$. [2]

   (d) A second line L_2 is perpendicular to L and passes through the point A. Find the equation of L_2. [3]

2. The graph of $f(x) = x^2 - 6x + 5$ is a parabola.

   (a) Find the x-intercepts of the graph of f. [2]

   (b) Write $f(x)$ in the form $(x - h)^2 + k$. [2]

   (c) Hence write down the coordinates of the vertex of the parabola. [1]

   (d) The graph of f is translated by the vector $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$. Find the equation of the new graph in the form $y = x^2 + bx + c$. [4]

3. The line $y = 2x + k$ is tangent to the parabola $y = x^2 - 4x + 7$.

   (a) Write down an equation whose solutions are the x-coordinates of the points of intersection of the line and the parabola. [2]

   (b) Hence show that $x^2 - 6x + (7 - k) = 0$. [1]

   (c) State the condition on the discriminant for the line to be tangent to the parabola. [1]

   (d) Find the value of k. [3]

   (e) Find the coordinates of the point of tangency. [3]

4. A ball is thrown vertically upward. Its height h metres above the ground at time t seconds is modelled by

$$h(t) = -5t^2 + 20t + 1, \quad \text{for } t \geq 0.$$

The velocity of the ball at time t is given by the gradient of the tangent to the curve $h(t)$. For a quadratic $h(t) = at^2 + bt + c$, the gradient of the tangent at t is $h'(t) = 2at + b$.

(a) Write down the height of the ball when $t = 0$. [1]

(b) Find $h'(t)$, the velocity of the ball at time t. [2]

(c) Find the time when the ball reaches its maximum height. [2]

(d) Find the maximum height. [2]

(e) Find the time when the ball hits the ground. [2]